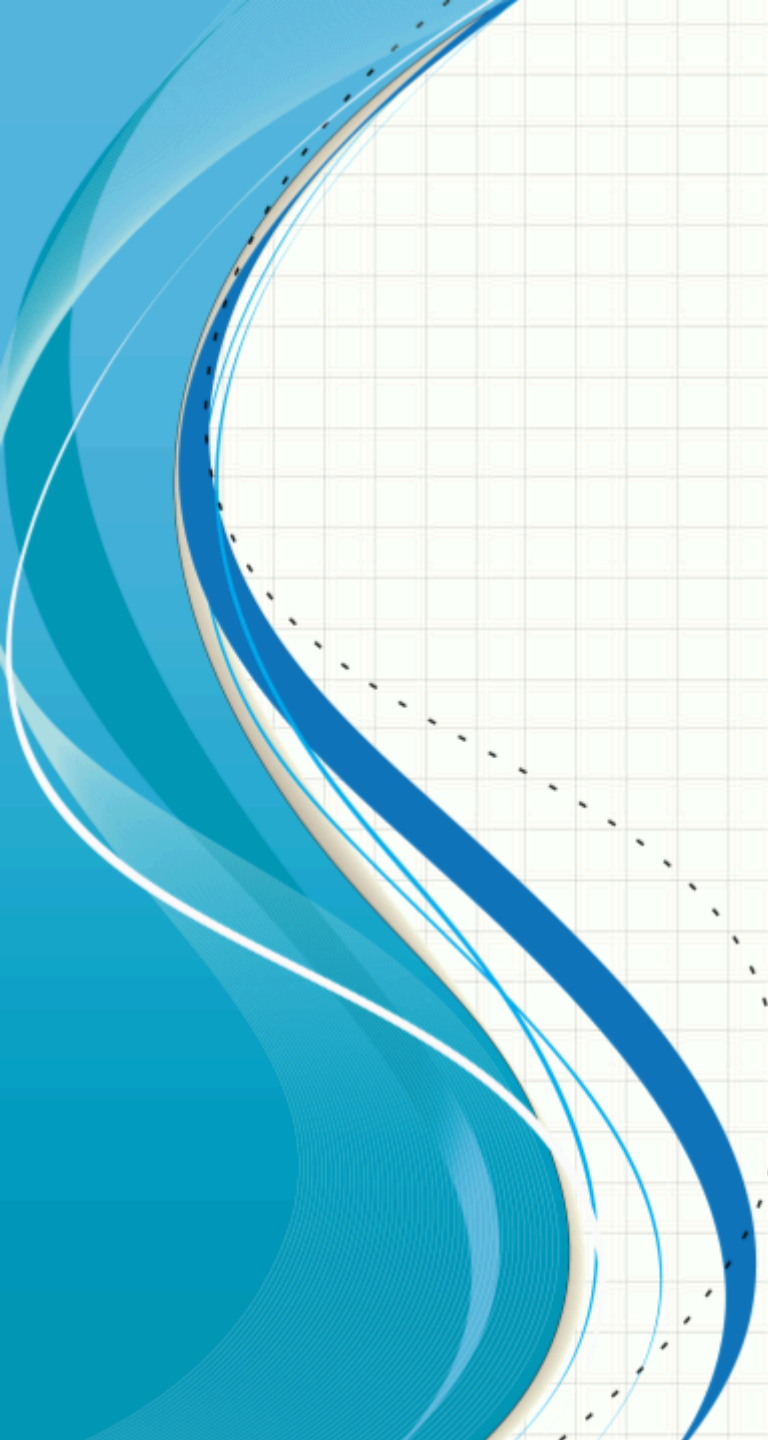




Pelvic inflammatory disease

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MRCOG, FICMS, MBChB

PID

is upper genital tract infection include:

Endometritis (endometrial infection).

Salpingitis (fallopian tube infection).

Oophoritis (infection of ovaries).

Parametritis (parametrial infection

PID

Fallopian tube

3. Bacteria enter fallopian tubes and ovaries, which can become infected

Uterus

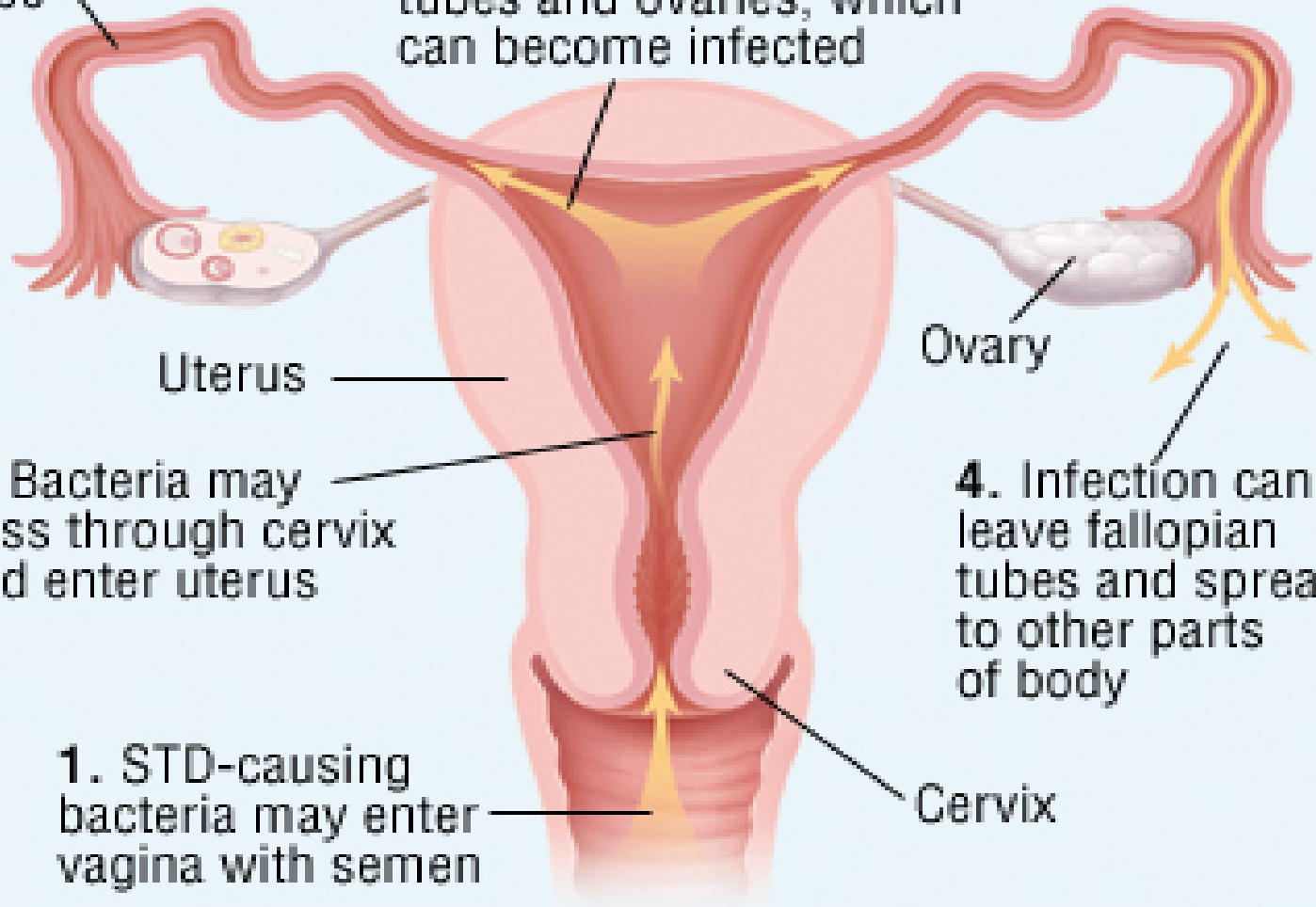
Ovary

2. Bacteria may pass through cervix and enter uterus

4. Infection can leave fallopian tubes and spread to other parts of body

1. STD-causing bacteria may enter vagina with semen

Cervix



vagina with semen
bacteria may enter

ovary

Risk factors for PID

- .young age (<25 yr.)
- .multiple sexual partner & recent sexual partner.
- .lack of condom use.
- .lower socio economic status.
- .past history of STD.
- .termination of pregnancy.
- .IUCD insertion in past 6 weeks.
- .HSG .IVF .bacterial vaginosis . Smoking .
- .post partum Endometritis .

MICROBIOLOGY

PID is a polymicrobial infection

Chlamydia trachomatis **30%**

Neisseria gonorrhea **5%**

Urea plasma, mycoplasma, gardnerella,

Anaerobes: Bactriod, peptostrepto
cocci, clostridium, fusobacterum.

Viruses: HSV , Echo virus, coxsacke virus

Clinical features:

Essential features

Lower abdominal pain.

Adnexal tenderness.

Cervical motion tenderness.

supporting features

intermenstrual bleeding,
PCB,

Vaginal discharge ,
deep dyspareunia, fever ,
nausea, vomiting,
generalized peritonitis.



Fitz-Hugh Curtis syndrome

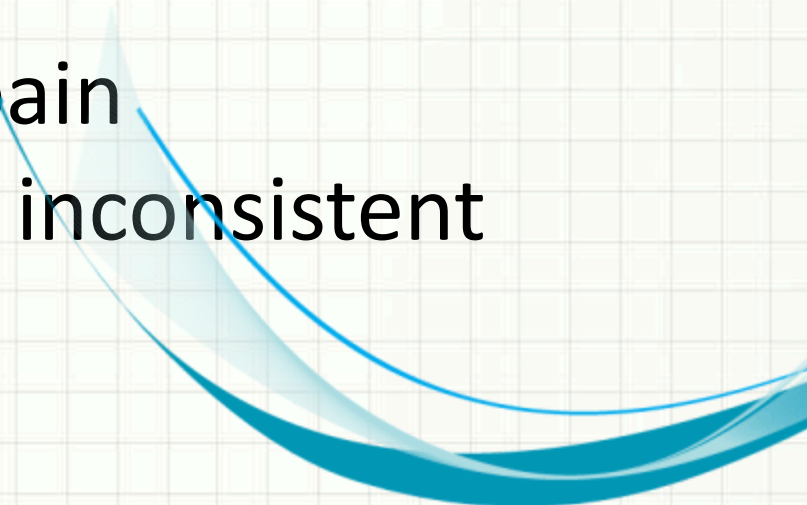
is inflammation & infection of liver capsule
(**perihepatitis**) affect
10-20% of female
with PID ,patient
present with Rt
upper abdominal
pain &tenderness.



Differential diagnosis :

1. Ectopic pregnancy :menstrual hx, unilateral
2. Ovarian accident: unilateral, often mid cycle
3. Appendicitis :GIT symptom, Rt side pain
4. IBS :central or Lt side abd.pain
5. UTI :urinary symp.+_ loin pain
(chlamydia can present with UTI)

Differential diagnosis :

- 6. Bowel torsion: central pain
 - 7. Psycho somatic: usually inconsistent pain
 - 8. Endometriosis .
 - 9. constipation.
- 

Laboratory Investigations of PID

- Pregnancy test
- Triple and urethral swabs
 - High vaginal swab – Bacteria vaginosis organisms
 - Endocervical swab – Neisseria gonorrhoea
 - Endocervical swab - Chlamydia trachomatis
 - Urethral swab – Chlamydia trachomatis (males only)
- Midstream Urine
 - Leucocytes and nitrates
- C-Reactive Protein and ESR
 - Markers for acute infection / inflammation
- The presence of leucocytes on vaginal wet preparation.

investigation:

1. **Pregnancy test** by blood hCG is **mandatory**

2. **Bl.test** :WBC

Raised white cell count (neutrophilia
suggestive of
acute inflammatory process)

Reduced white cell count (neutropenia in
severe infections)

ESR,C-reactive protein are **non** specific .

investigation:

3. Microbiological test:

- . Endocervical swab for gonorrhea culture (need transport media Stuarts or Amie's media)
- . Endocervical swab for chlamydia NAAT or EIA test **lack** sensitivity
- . Screening for other STD(HIV Ab test, Syphilis serology, Trichomonas vaginalis sample)

Radiological investigation

1. TV U/S its help full to exclude ectopic pregnancy, Ov.cyst, appendicitis, dilated tubes or tubal abscess.

2. Power Doppler has little benefit

3. MRI, CT not routinely use



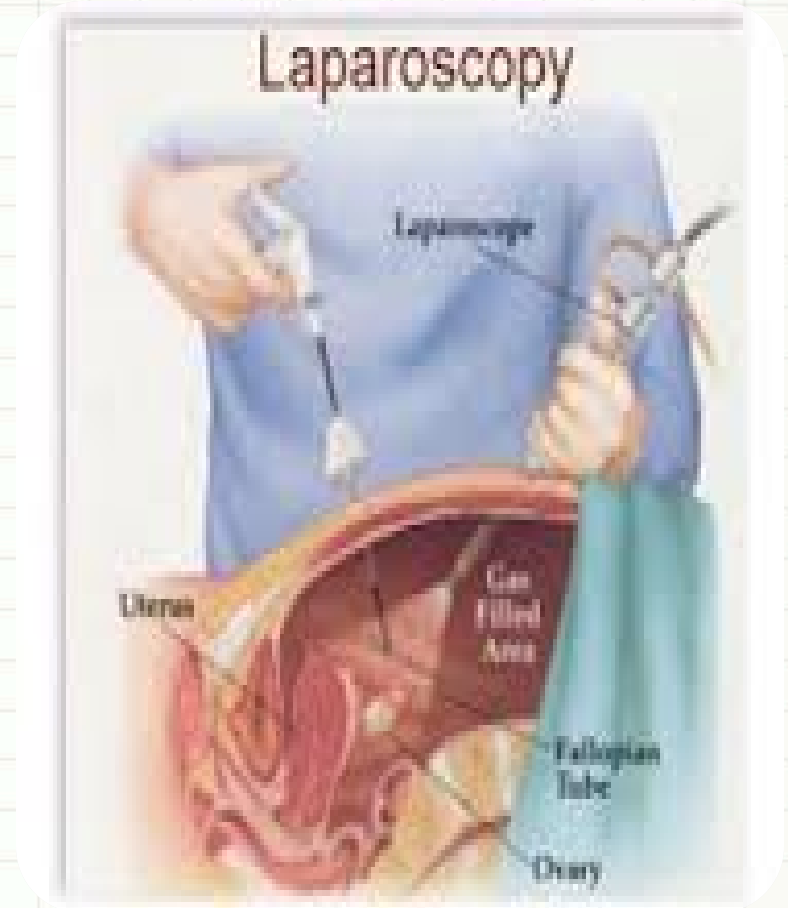
Surgical investigation

1.laparoscopy is invasive
use in case of

If there is doubt to the
diagnosis .

Patient **fail** to respond to
AB within 48-72 hr.

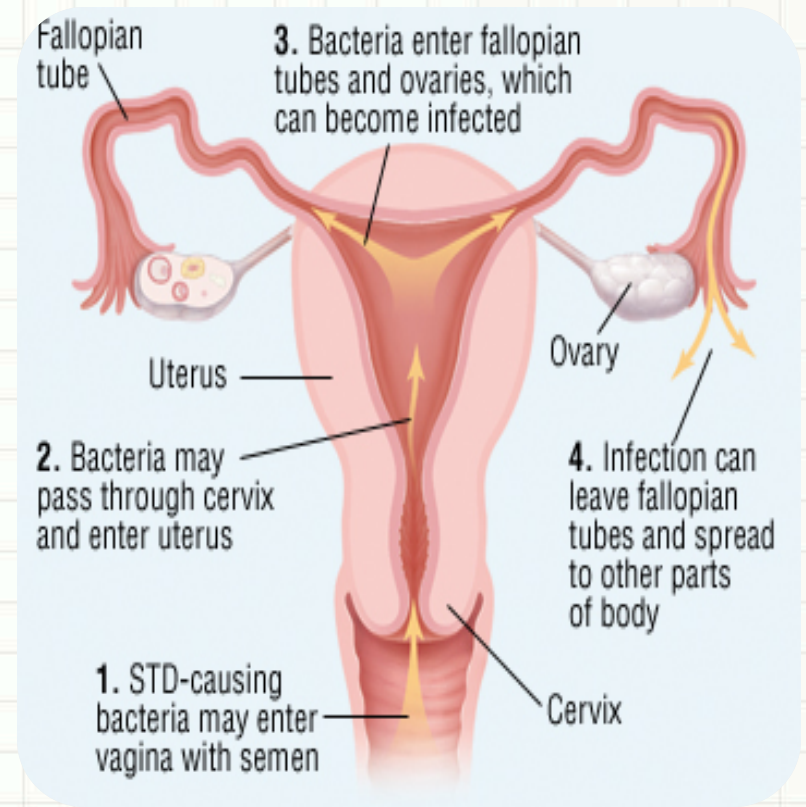
2.Hysteroscopy is **not**
routine invx.



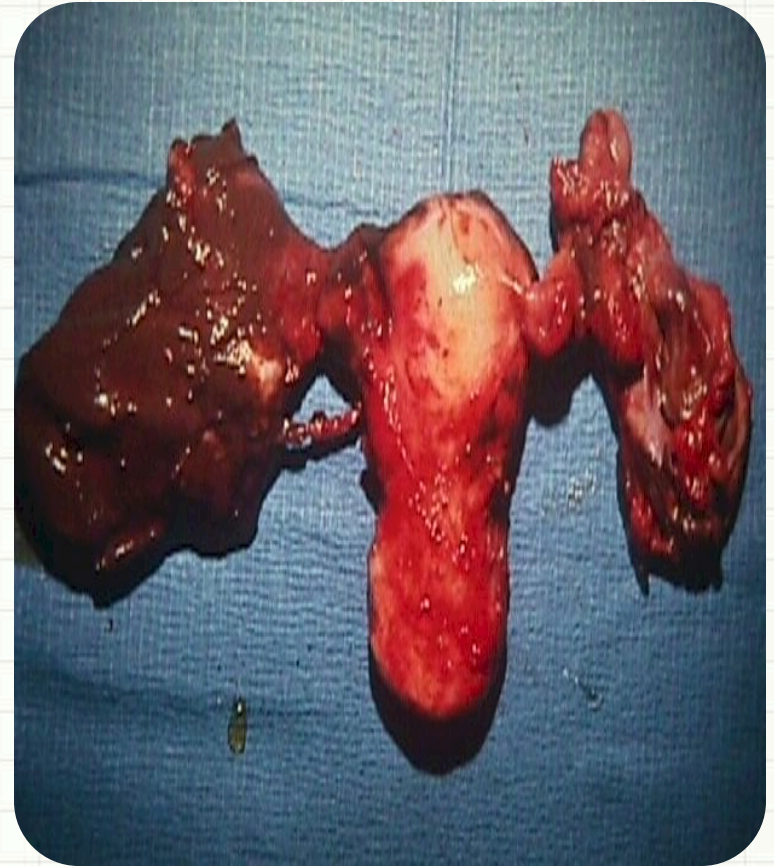
Histology & pathology

the spread of infection from cervix to endometrium lead to acute **Polymorph** mediated Endometritis.

Gonorrhea in tube affect non ciliated ep.cell but produce **TNF** & **gama IF** lead to damage of tissue & invasion to sub mucosa



chlamydia cause tissue damage by immune response (delayed type hyper sensitivity reaction)
Reinfection with chlamydia lead to further immune stimulation & tissue damage.



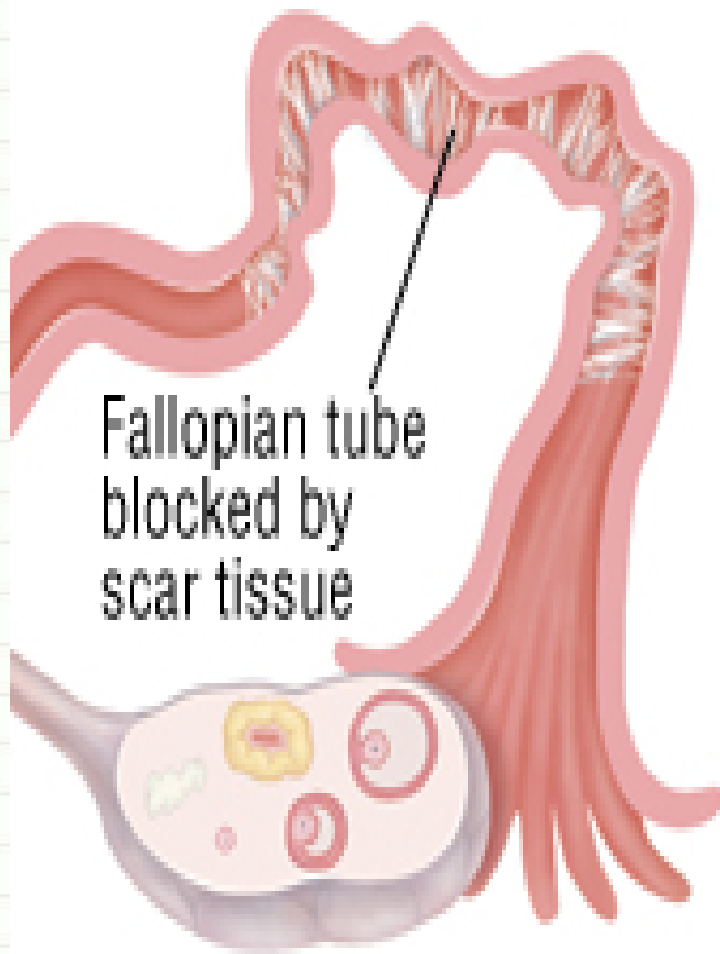
Complication of PID

Sever inflammation is associated with:

1. Chronic pelvic pain due to recurrent infection & adhesion

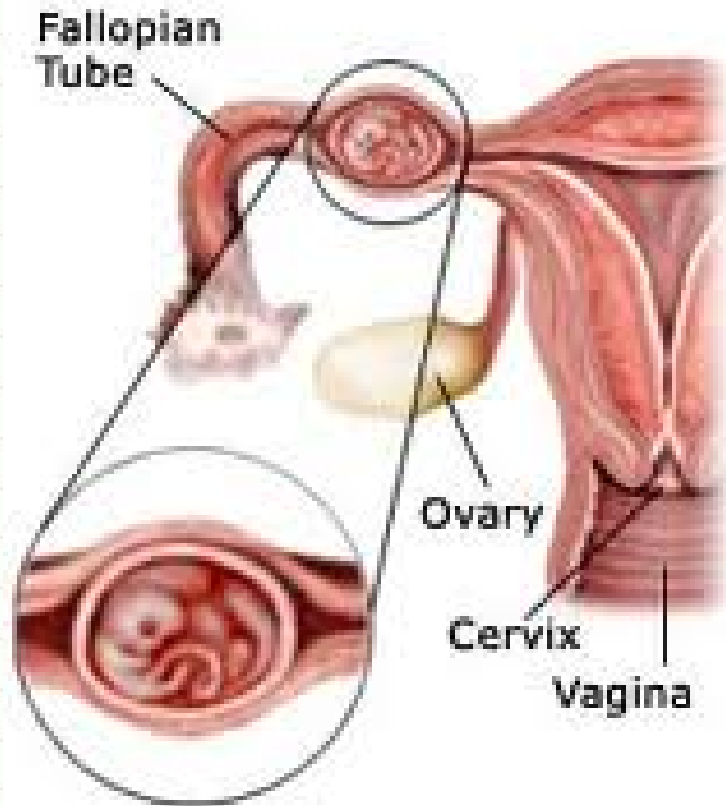
Pelvic nerve damage it affect 33% of cases after recurrent infection





2. Sub fertility :tubal occlusion & tubo ovarian abscess or hydro salpenx: healing lead to chronic fibrosis& damage to ciliated ep. Lead to tubal blockage.

Etopic Pregnancy



3. ectopic pregnancy

Damage to ciliated epithelium lead to ectopic pregnancy

Treatment

- *Principles of treatment*
- • Adequate supportive care
- • Strict watch on fluid balance
- • Parenteral antibiotics

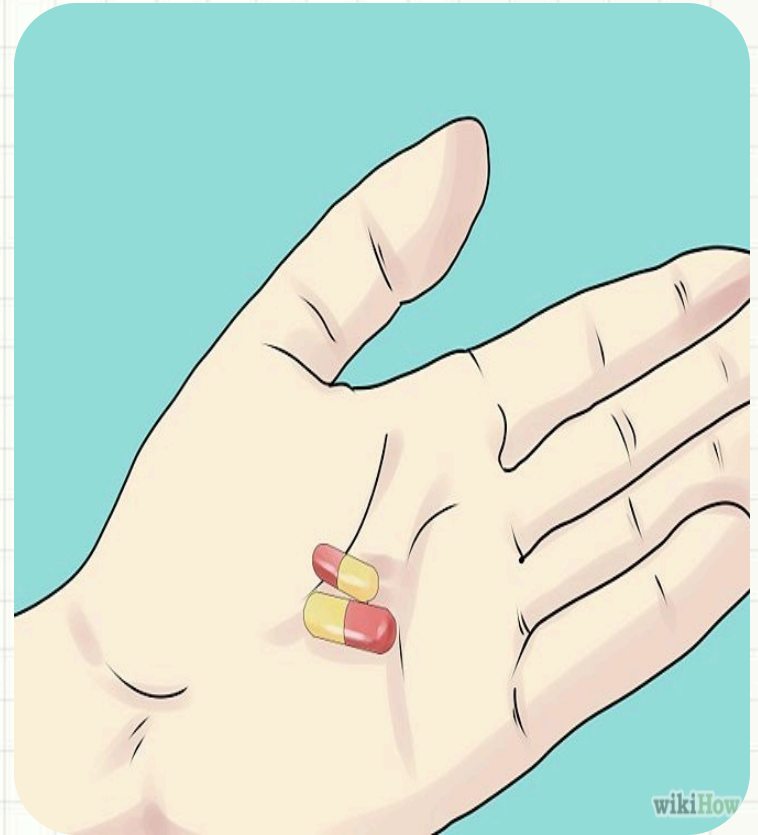
1. general measures

- . Rest for severe disease
- . appropriate analgesia
- . pregnancy test
- . inpatient management for severe cases
- . avoid sexual contact
- . full explanation
- . STD screening



Antibiotics

Broad spectrum AB. To cover gonorrhea, chlamydia & anaerobes.
Optimal AB. Depend on
.local bacterial resistance.
.severity of disease
.cost
.patient convenience



Antibiotics

- Intravenous therapy should be continued until 24 hours after clinical improvement
- Inpatient regimens
 - ● ceftriaxone 2 g by intravenous infusion daily plus intravenous doxycycline 100 mg twice daily, followed by oral doxycycline 100 mg twice daily plus oral metronidazole 400 mg twice daily for a total of 14 days;
 - ● oral doxycycline may be used, if tolerated, while on intravenous ceftriaxone.
 - or
 - ● intravenous clindamycin 900 mg three times daily plus intravenous gentamicin, followed by either – oral clindamycin 450 mg four times daily to complete 14 days.
 - or

Antibiotics

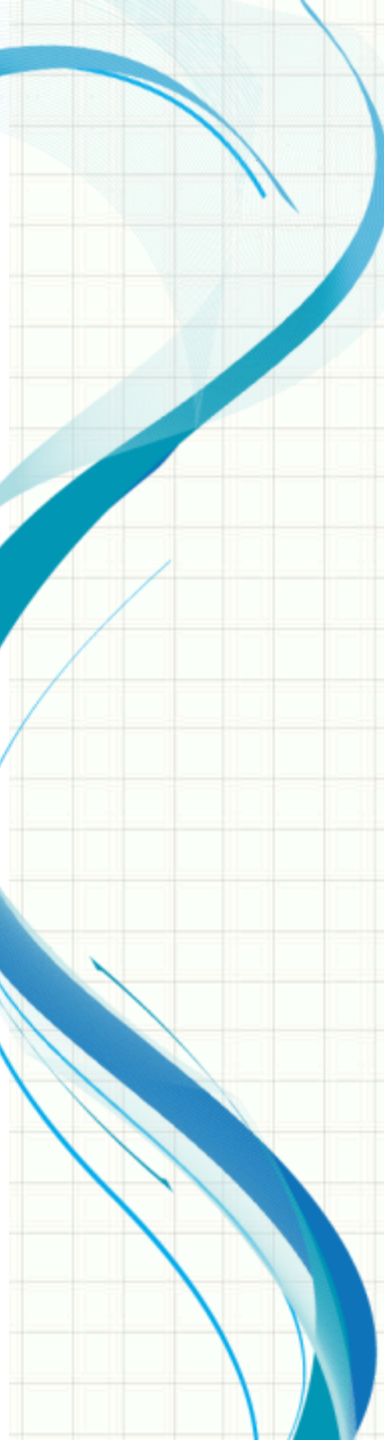
- or
- – oral doxycycline 100 mg twice daily plus oral metronidazole 400 mg twice daily to complete 14 days.
- ● Gentamicin should be given as a 2 mg/kg loading dose
- followed by 1.5 mg/kg three times daily (or a single daily
- dose of 7 mg/kg may be substituted).
- ● Intravenous ofloxacin 400 mg twice daily plus intravenous metronidazole 500 mg three times daily for 14 days.

Antibiotics

- Outpatient regimens
 - Oral ofloxacin 400 mg twice daily plus oral metronidazole 400 mg twice daily for 14 days.
 - Intramuscular (i.m.) ceftriaxone 250 mg single dose or i.m. cefoxitin 2 g single dose with oral probenecid 1 g followed by oral doxycycline 100 mg daily plus metronidazole 400 mg twice daily for 14 days.
 - Ofloxacin should be avoided in women at high risk of gonococcal PID due to increased quinolone resistance in the United Kingdom..

current male partner
should be offered
screening for STD.
& attempt made to
contact other
partners within past
6months



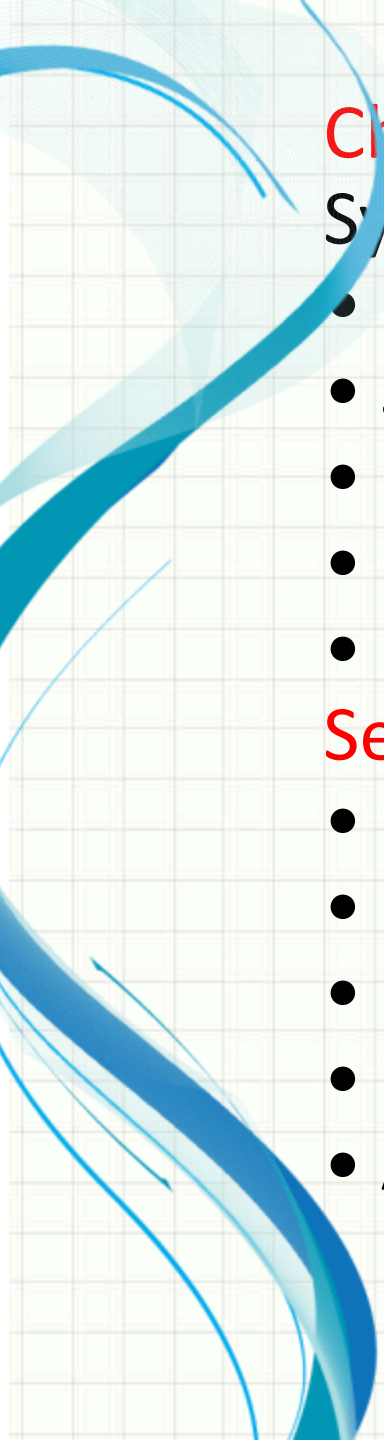


surgical intervention:

.To drain a pelvic abscess if this diagnosed on U/S & not resolve with AB treatment

Most surgeon prefer laparotomy

. In case of small abscess or fluid collection in the pouch of Douglas U/S guided aspiration is less invasive.



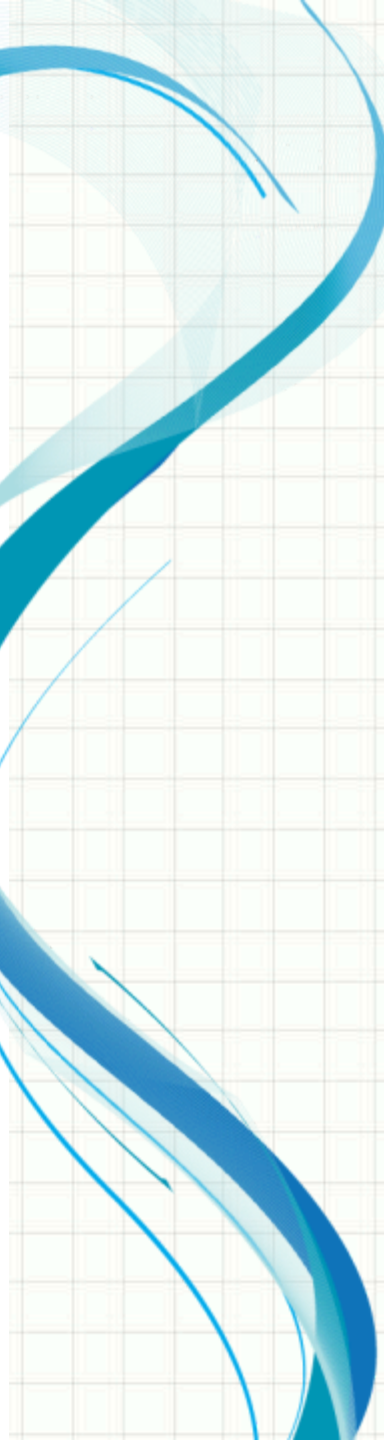
Chronic Pelvic Inflammatory Disease

Symptoms > 6 Months Duration

- Pelvic pain
- Secondary dysmenorrhoea
- Deep dyspareunia
- Menstrual disturbance
- Recurrent acute painful exacerbations

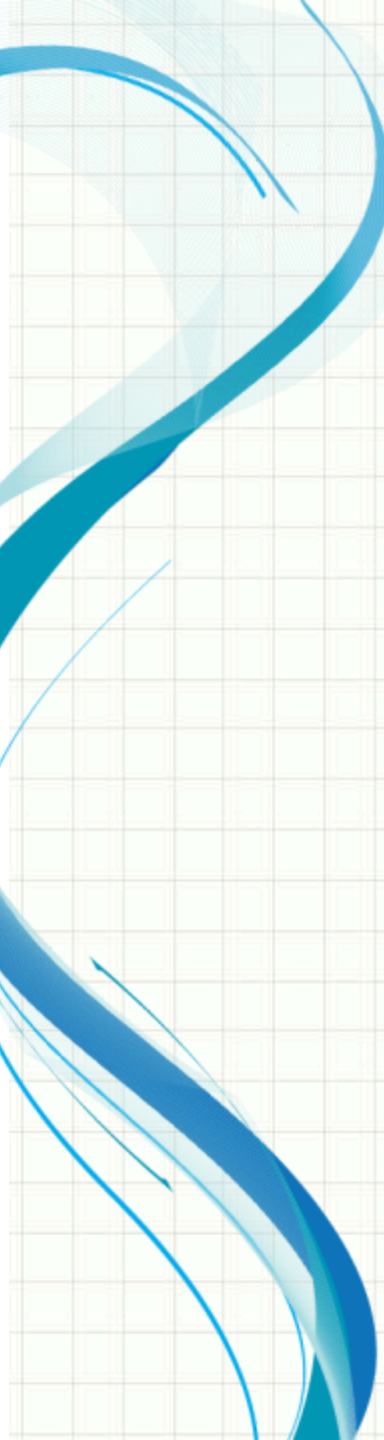
Sequelae

- Infertility
- Ectopic pregnancy
- Chronic pelvic pain (pelvic cripple)
- Pelvic adhesions/ tubo-ovarian complex
- Abnormal / painful periods

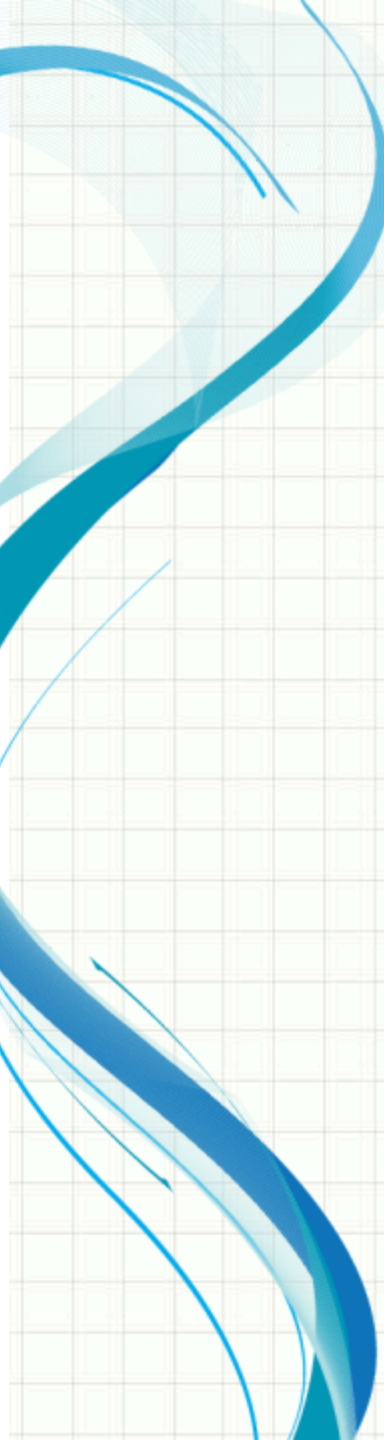


case history

A 29-year-old woman presents to the emergency gynaecology unit with a 5-day history of lower abdominal and pelvic pain which has got progressively worse over the last 24 hours. She also feels generally unwell with chills and rigors.



what are other
important point
in history?



1. Her LMP

2. Gynaecological history :

menstrual cycle

PCB

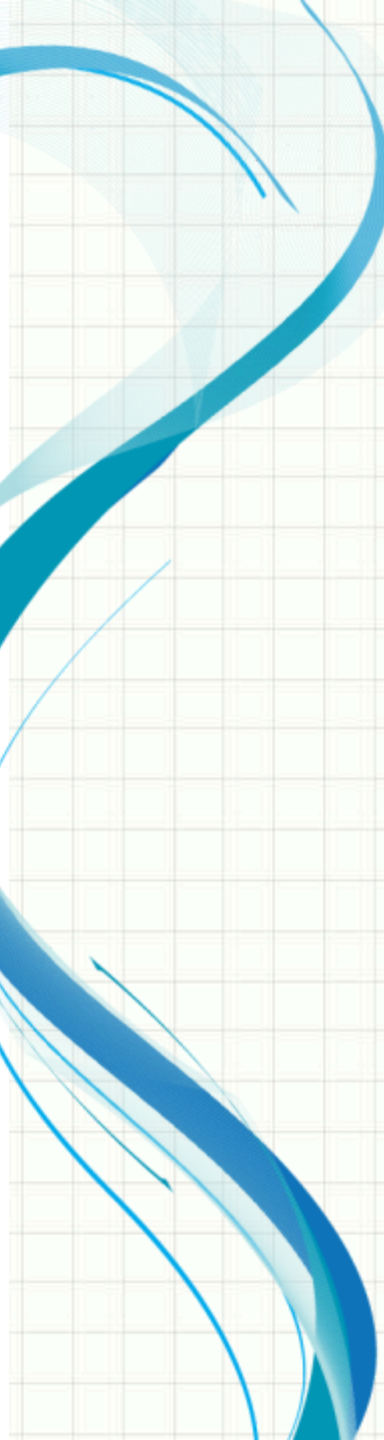
vaginal discharge

contraception

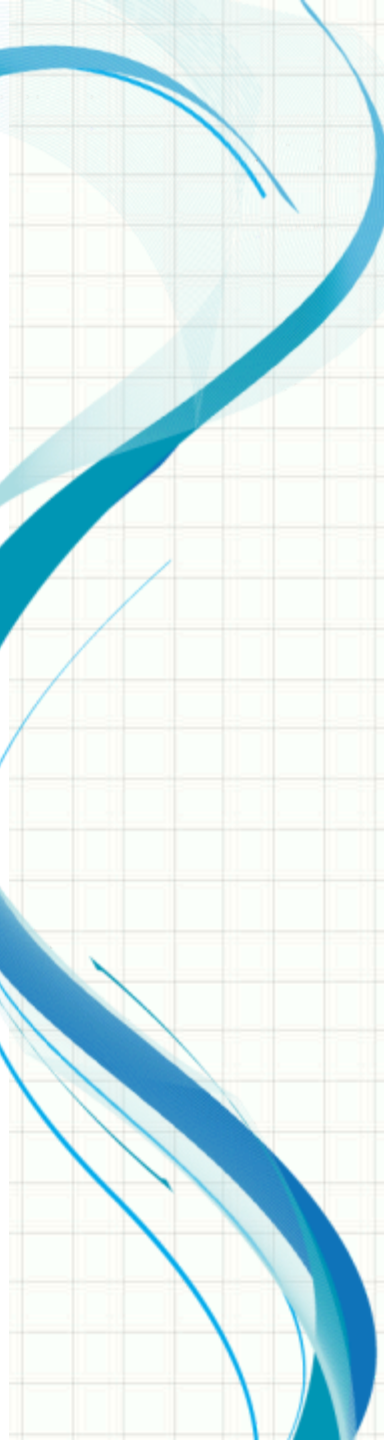
past history of PID

history of surgical maneuver

3. Smoking

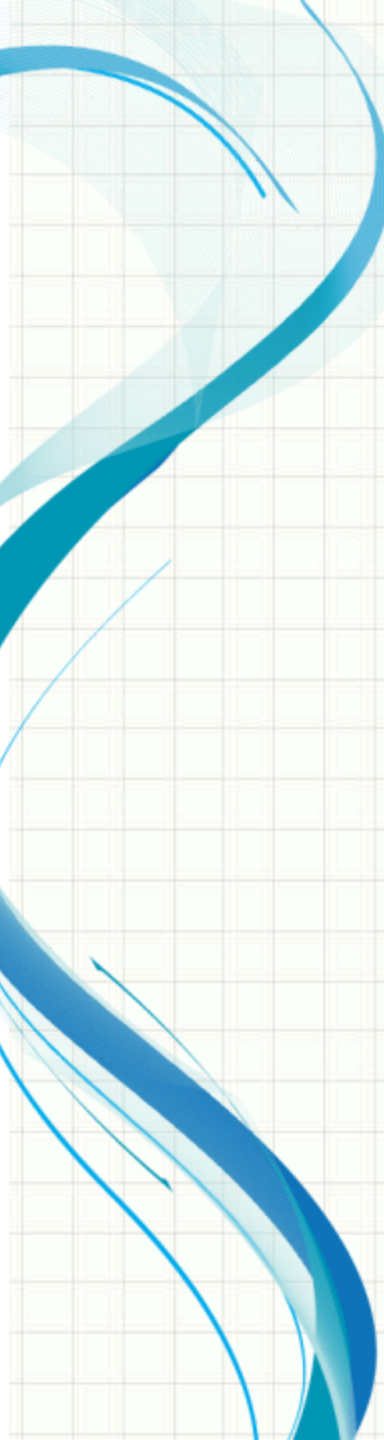


what are important point in
examination ?

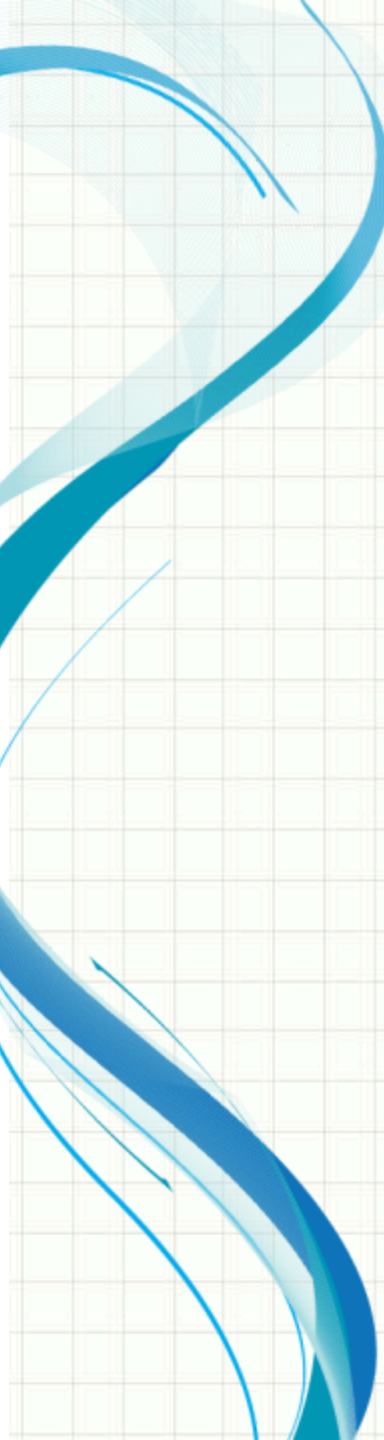


what are important point in examination :

1. General examination: she looks very unwell.
2. Temperature: She has a raised temperature at 38°C and is tachycardia.
3. Abd. exam: She is very tender on abdominal examination with guarding.
4. On speculum examination, the cervix appears inflamed with profuse mucopurulent discharge.
5. A high vaginal swab, endocervical swab and a urethral swab are taken.
6. On pelvic examination, there is marked cervical tenderness. She is also generally tender to palpate throughout the pelvis and is unable to tolerate the examination.
7. A pregnancy test is negative.

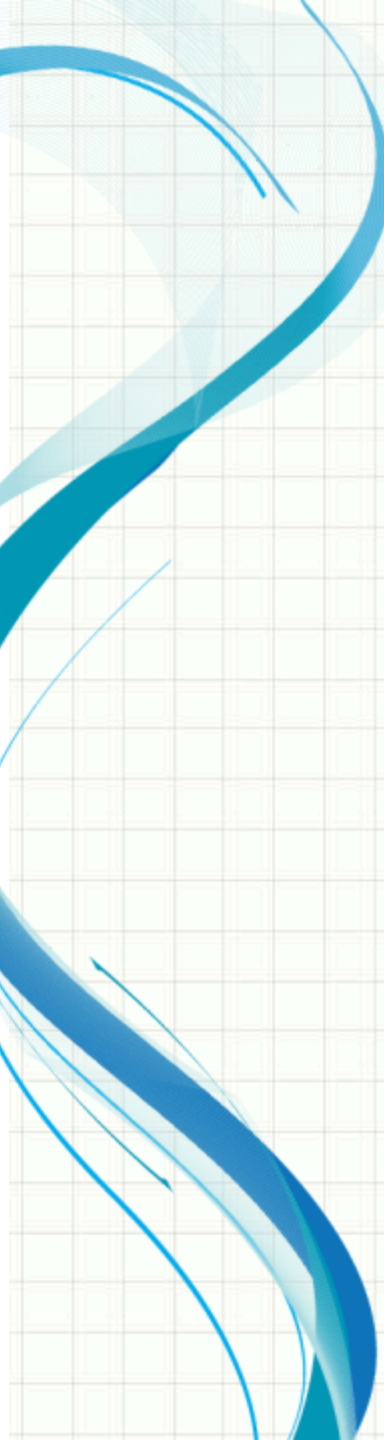


what is your diagnosis ?



what is your diagnosis ?

The clinical diagnosis is suggestive of acute pelvic inflammatory disease with pelvic peritonitis

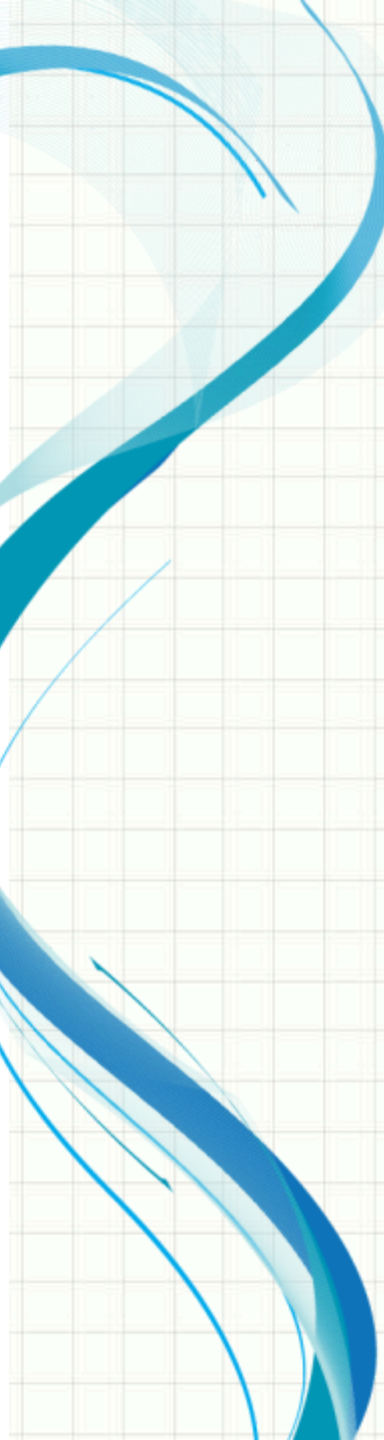


what are the step in treatment ?

General measure & antibiotic

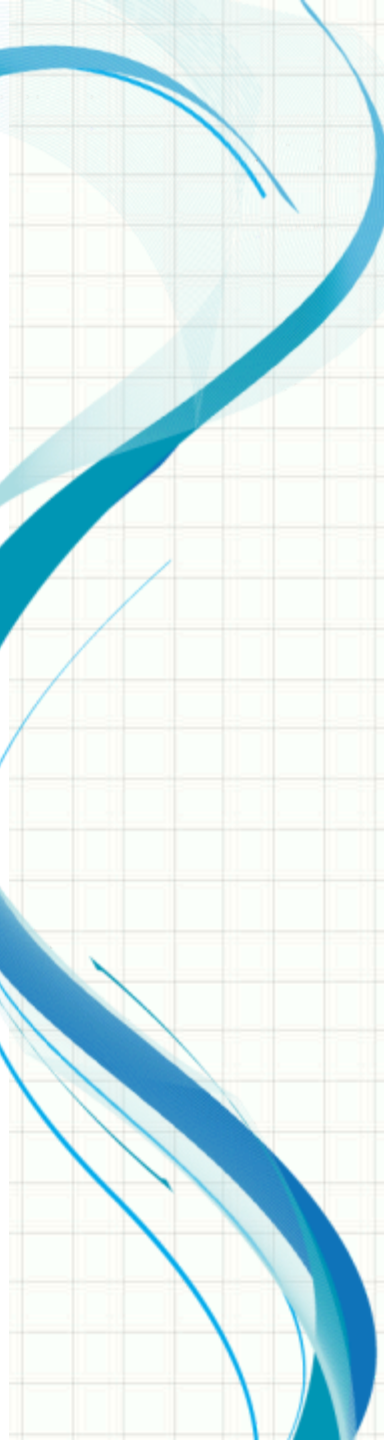
She is admitted and started with intravenous antibiotics (ceftriaxone + metronidazole) and oral doxycycline.

She is also supported with intravenous fluids, analgesics and regular paracetamol.



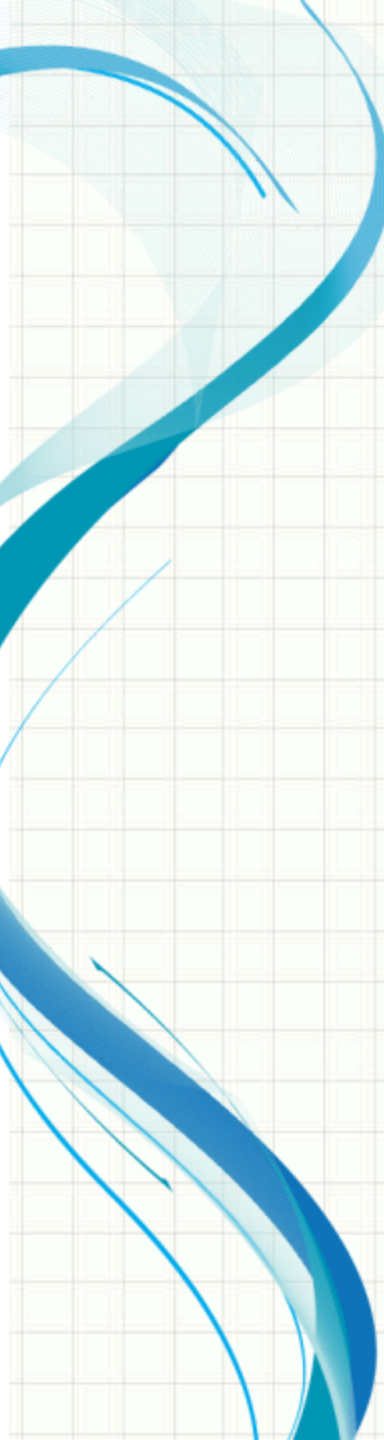
She continues to be unwell with a raised temperature even after 24 hours of antibiotics.

What is the next step ??



A pelvic ultrasound
It shows the possibility of an adnexal
mass with some free fluid in the pelvis.

What is the next step ??



a decision is taken to perform a laparoscopy.

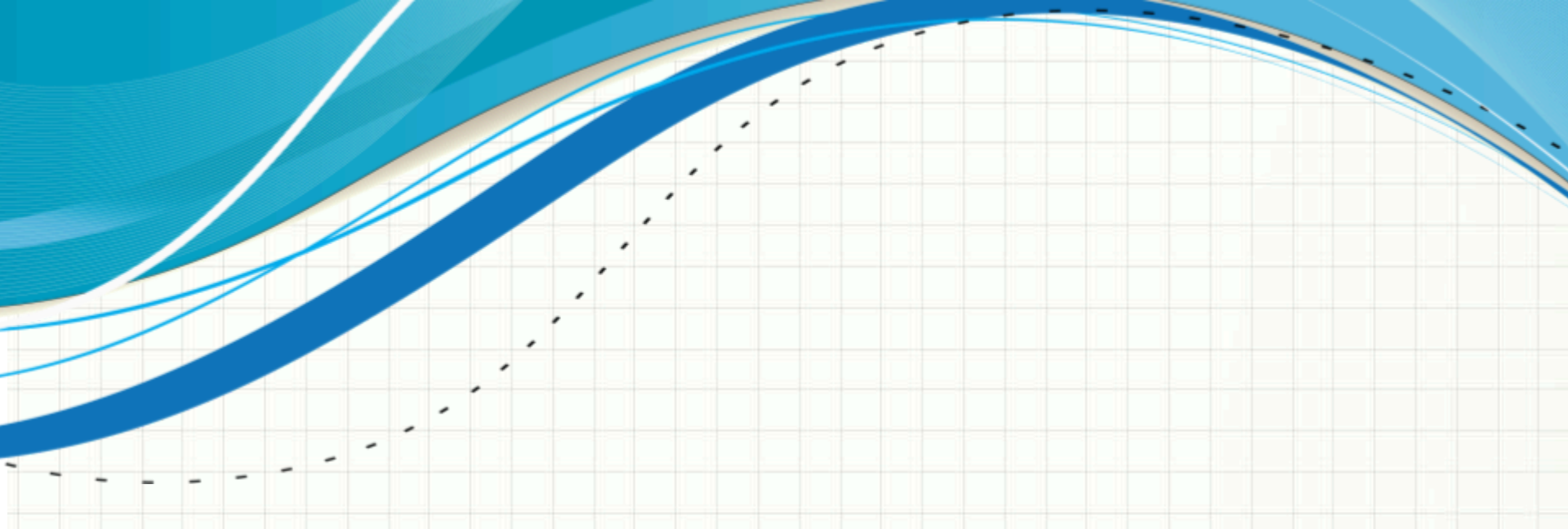
At the operation, she is found to have a 7-cm enlarged tubo-ovarian abscess and marked inflammation of the uterus and other tube.

The abscess is drained and the patient markedly improves after the surgery. She is found to have chlamydia on the endocervical swab with anaerobes on the high vaginal swab. She improves over the course of a few days and is treated with oral antibiotics **for 2 weeks**.

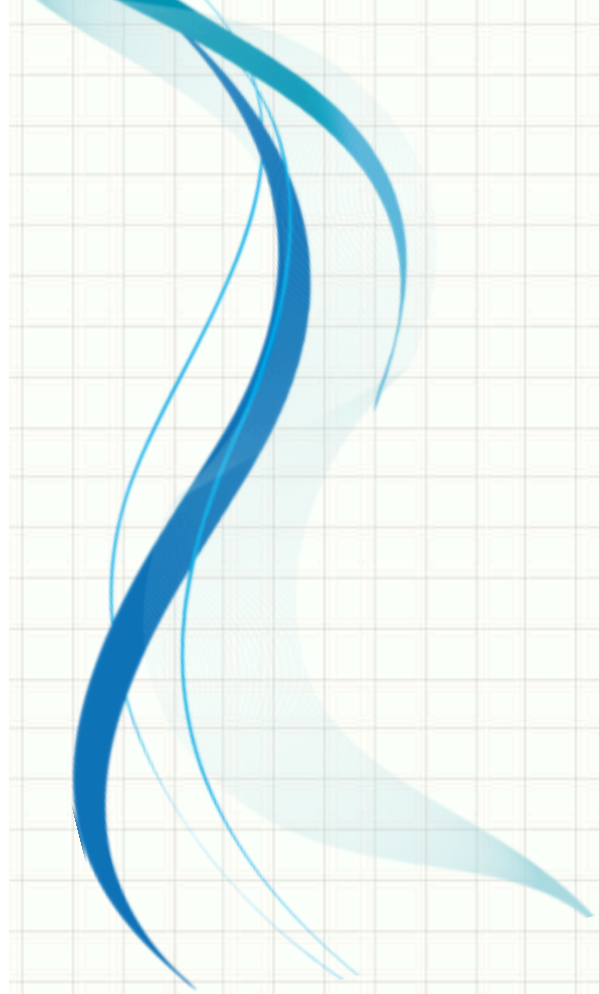
What is the next step ???



She is counselled regarding the implications of the infection and is encouraged to contact her present and previous partners for testing, of whom one tested positive for *Chlamydia* and was subsequently treated .



Questions?



Welcome